

U.G.C. NET EX COMPUTER (Solved with Expla

1. **The Software Maturity Index (SMI) is defined as $SMI = [M_t - (F_a + F_c + F_d)] / M_t$ Where**
 M_t = the number of modules in the current release.
 F_a = the number of modules in the current release that have been added.
 F_c = the number of modules in the current release that have been changed.
 F_d = the number of modules in the current release that have been deleted.
The product begins to stabilize when
 (a) SMI approaches 1
 (b) SMI approaches 0
 (c) SMI approaches -1
 (d) None of the above

Ans : (a) Software maturity index: Based on number of modules in current release and number of modules changed, added, deleted in current release:

$$SMI = [M_t - (F_a + F_c + F_d)] / M_t$$

where,

M_t = # modules in current release

F_c = modules changed in current release

F_a = # modules added

F_d = # modules deleted

$$SMI = [M_t - (0 + 0 + 0)] / M_t = 1 \text{ (good)}$$

The product begins to stabilized when no new module is added in current release (i.e., $F_a = 0$), no change happen in old release, (i.e., $F_c = 0$), no module is removed from old release ($F_d = 0$).

2. **Match the following:**

List-I

A. Watson-Felix model

B. Quick-Fix model

C. Putnam resource allocation model

D. Logarithmic-Poisson Model

List-II

1. Failure intensity

2. Cost estimation

3. Project planning

4. Maintenance

Codes:

	A	B	C	D
(a)	2	1	4	3
(b)	1	2	4	3
(c)	2	1	3	4
(d)	2	4	3	1

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Ans : (d) Watson-Felix model is a cost estimation model similar to cost estimation which says effort, $E = 5.2(KLOC)^{0.91}$.

Quick fix model is a maintenance approach in which we wait till problem occurs and fix it as quickly as possible.

Putnam resource allocation model is used for effort estimation for a project and is used while planning a project.

A logarithmic Poisson execution time model is for software reliability measurement.

3. ____ is a process model that removes defects before they can precipitate serious hazards.
- (a) Incremental model
 - (b) Spiral model
 - (c) Cleanroom software engineering
 - (d) Agile model

Ans : (c) Clean Room Software Engineering (CSE) is a process model that removes defects before they can precipitate serious hazards. It is a team oriented, theory based software, which is developed using the formal methods, correctness verification and Statistical Quality Assurance (SQA). Clean room management is based on the incremental model of software development, which accumulates into the final product.

The goal of clean room-engineering is to produce zero error based software by allowing correct designs.

4. **Equivalence partitioning** is a _____ method that divides the input domain of a program into classes of data from which test cases can be derived.
- (a) White-box testing
 - (b) Black-box testing
 - (c) Orthogonal array testing
 - (d) Stress testing

Ans : (b) The general tendency is to relate **equivalence partitioning** to **black box testing** which is strictly checking a software component at its interface, without any consideration of internal structures of the software.

5. **The following three golden rules:**
- (i) **Place the user in control**
 - (ii) **Reduce the user's memory load**
 - (iii) **Make the interface consistent**
- are for :**
- (a) User satisfaction
 - (b) Good interface design
 - (c) Saving system's resources
 - (d) None of these

Ans : (b) There are certain universal golden rules of user interface design regardless of the domain, user interface, or intended device (Computer, Tablet or Phone)

These are:

1. Place users in control
2. Reduce users memory load
3. Make the interface consistent

Each of these groups contains a number of specific rules.

6. **Software safety is a _____ activity that focuses on the identification and assessment of potential hazards that may affect software negatively and cause an entire system to fail.**
- (a) Risk mitigation, monitoring and management
 - (b) Software quality assurance
 - (c) Software cost estimation
 - (d) Defect removal efficiency

Ans : (b) **Software safety** is a software quality assurance activity that focuses on the identification and assessment of potential hazards that may affect software negatively and cause an entering system to fail.

Examples of possible failures are:

1. Uncontrolled acceleration of a vehicle.
2. Locked doors.
3. Prevention braking.

7. **The "PROJECT" operator of a relational algebra creates a new table that has always**
- (a) More columns than columns in original table
 - (b) More rows than original table
 - (c) Same number of rows as the original table
 - (d) Same number of columns as the original table

Ans : (*) Project in relation algebra remove duplicate. The project operation is unary operation. This operation returns a set of tuples containing a subset of the attributes in the original relation. It discards the duplicate rows.

So, none option is true. Note that official key is given option (c) which is wrong.

8. **The employee information of an Organization is stored in the relation:**
Employee (name, sex, salary, deptname)
Consider the following SQL query
Select deptname from Employee
Where Sex ='M'
Group by deptname
Having avg (salary)>
{select avg (salary)from Employee}
Output of the given query corresponds to
- (a) Average salary of employee more than average salary of the organization
 - (b) Average salary less than average salary of the organization
 - (c) Average salary of employee equal to average salary of the organization.
 - (d) Average salary of male employees in a department is more than average salary of the organization.

Ans : (d) Select deptname
 From employee
 Where sex = 'M'
 group by dept name
 having avg (salary) > (select avg (salary) from employee)
 We are selecting male employee by grouping all the male employee of a department whose average salary is greater than the average salary of organization.

- 9. For a database relation R (a,b,c,d) where the domains of a, b, c, d include only the atomic values. The functional dependency $a \rightarrow c$, $b \rightarrow d$ holds in the following relation**
- (a) In 1NF not in 2 NF
 - (b) In 2 NF not in 3 NF
 - (c) In 3 NF
 - (d) In 1 NF

Ans : (a) Given relation is $R(a, b, c, d)$ with functional dependencies:
 $\{a \rightarrow c, b \rightarrow d\}$
 So candidate key is $\{ab\}$.
 The given relation has partial dependency (both $a \rightarrow c$, and $b \rightarrow d$). Such kind of dependencies (are not allowed in 2NF.
 So, relation R is not in 2NF, but is in 1NF.

- 10. Match the following:**

List-I	List-II
A. RAID 0	1. Bit interleaved parity
B. RAID 1	2. Non redundant stripping
C. RAID 2	3. Mirrored disks
D. RAID 3	4. Error correcting codes

Codes:

	A	B	C	D
(a)	4	1	2	3
(b)	3	4	1	2
(c)	3	1	4	2
(d)	2	3	4	1

Ans : (d) **1. RAID 0** (also known as a stripe set or striped volume) splits data evenly across two or more disks, without parity information redundancy, or fault tolerance.
2. RAID 1 consists of an exact copy (or mirror) of a set of data on two or more disks: a classic RAID 1 mirrored pair contains two disks.
3. RAID 2, which is rarely used in practice, stripes data at the bit (rather than block) level, and uses a Hamming code for error correction.
4. RAID 3, which is rarely used in practice, consist of byte-level striping with a dedicated parity disk.

- 11. The golden ratio ϕ both satisfy the equation**

- (a) $x^3 - x - 1 = 0$
- (b) $x^3 + x - 1 = 0$
- (c) $x^2 - x - 1 = 0$
- (d) $x^3 + x - 1 = 0$

Ans : (c) $\phi^2 - \phi - 1 = \left(\frac{1+\sqrt{5}}{2}\right)^2 - \left(\frac{1+\sqrt{5}}{2}\right) - 1 = 0$

And now the conjugate:

$$\hat{\phi}_2 - \hat{\phi} - 1 = \left(\frac{1-\sqrt{5}}{2}\right)^2 - \left(\frac{1-\sqrt{5}}{2}\right) - 1 = 0$$

Therefore, the golden ratio ϕ and its conjugate $\hat{\phi}$ both satisfy the equation

$$x^2 - x - 1 = 0$$

12. The solution of recurrence relation,

$$T(n) = 2T(\lfloor \sqrt{n} \rfloor) + \log n$$

- (a) $O(n \log \log \log n)$
- (b) $O(n \log \log n)$
- (c) $O(\log \log n)$
- (d) $O(\log \log \log n)$

Ans : (d) Given recurrence relation is:

$$T(n) = 2T(\lfloor \sqrt{n} \rfloor) + \log n$$

Substituting

$$n = 2^m$$

$$\Rightarrow m = \log n$$

$$T(2^m) = 2T(2^{m/2}) + m$$

We can rename $S(m) = T(2^m)$ to produce the new recurrence

$$S(m) = 2S(m/2) + m$$

$$a = 2, b = 2, f(m) = m$$

$$O(m^{\log_2(2)}) = f(m) \text{ [case (2) master theorem]}$$

So, $T(n) = \Theta(m \log_2 m)$

replace $m = \log_2(n)$, therefore,

$$T(n) = \Theta(\log n \log \log n)$$

13. In any n-element heap, the number of nodes of height h is

- (a) less than equal to $\left\lceil \frac{n}{2^h} \right\rceil$
- (b) greater than $\left\lceil \frac{n}{2^h} \right\rceil$
- (c) greater than $\left\lceil \frac{n}{2^h + 1} \right\rceil$
- (d) less than equal to $\left\lceil \frac{n}{2^h + 1} \right\rceil$

Ans : (d) Theorem: The number of nodes at height h in a heap tree are almost $\left\lceil \frac{n}{2^{h+1}} \right\rceil$.

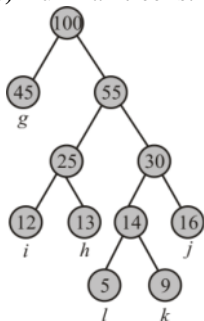
14. A data file of 1,00,000 characters contains only the characters g–I, with the frequencies as indicated in table:

	g	h	i	j	k	l
Frequency in thousand	45	13	12	16	9	5

using the variable-length code by Huffman codes the file can be encoded with.

- (a) 2,52,000 bits
- (b) 2,64,000 bits
- (c) 2,46,000 bits
- (d) 2,24,000 bits

Ans : (d) Huffman tree is:



So, total weight is:

Sum of internal node's value

$$= 100 + 55 + 45 + 30 + 14 = 224$$

$$= 224,000 \text{ bits}$$

15. A vertex cover of an undirected graph $G(V, E)$ is a subset $V_1 \subseteq V$ vertices such that

- (a) Each pair of vertices in V_1 is connected by an edge
- (b) If $(u, v) \in E$ then $u \in V_1$ and $v \in V_1$
- (c) If $(u, v) \in E$ then $u \in V_1$ or $v \in V_1$
- (d) All pairs of vertices in V_1 are not connected by an edge

Ans : (c) Vertex cover: Given an undirected graph $G = (V, E)$, a vertex cover is a subset of vertices $S \subseteq V$ such that for each edge $(U, V) \in E$, either $U \in S$ or $V \in S$ or both.

16. In a fully connected mesh network with n devices, there are _____ physical channels to link all devices.

- (a) $n(n-1)/2$
- (b) $n(n+1)/2$
- (c) $2n$
- (d) $2n+1$

Ans : (a) A fully connected network, complete topology, or full mesh topology is a network topology in which there is a direct link between all pairs of nodes. In a fully connected network with n nodes, there are $\frac{n(n-1)}{2}$ direct link.

17. The baud rate of a signal is 600 baud/second. If each signal unit carries 6 bits, then the bit rate of a signal is _____.

- (a) 3600
- (b) 100
- (c) 6/600
- (d) None of these

Ans : (a) Baud rate 600 baud/second means where 600 characters are transmitted in a second each character is of 6 bits.

Therefore, Bit rate = $600 \times 6 = 3600$ bps

18. Match the following :

List-I

- A. Data link layer
- B. Network layer
- C. Transport layer
- D. Application layer

List-II

- 1. Flow control
- 2. Node to node delivery
- 3. Mail services
- 4. Routing

Codes:

	A	B	C	D
(a)	2	1	4	3
(b)	2	4	1	3
(c)	2	1	3	4
(d)	2	4	3	1

Ans : (b) Data link layer is responsible for node (hop) to node (hop) delivery and transport layer is responsible for process (end) to process (end) delivery.

Routing is the process of construction of routing table. At the transport layer **flow control** will allow the transport protocol entity in a host to restrict the flow of data over a logical connection from the transport protocol entity to another host.

SMTP (Simple Mail Transfer Protocol) is an application layer protocol for electronic mail (email) transmission.

- 19. An image is 1024×800 pixels with 3 bytes/pixel. Assume the image is uncompressed. How long does it take to transmit it over a 10 Mbps Ethernet?**
- (a) 196.6 seconds
 (b) 19.66 seconds
 (c) 1.966 seconds
 (d) 0.1966 seconds

Ans : (c) Transmission time

$$\begin{aligned}
 &= \frac{\text{Data}}{\text{Bandwidth}} \\
 &= \frac{1024 \times 800 \times 3 \times 8}{10 \times 10^6} \\
 &= 1.96608 \text{ seconds}
 \end{aligned}$$

- 20. The _____ measures the relative strengths of two signals or a signal at two different points.**
- (a) frequency
 (b) attenuation
 (c) throughput
 (d) decibel

Ans : (d) To show that a signal has lost or gained strength, engineers use the concept of decibel. The **decibel (dB)** measures the relative strengths of two signals or a signal at two different points. Note that the dB is negative if a signal is attenuated and positive if a signal is amplified.

$$dB = 10 \log_{10} (P_1 / P_2)$$

Where P_1 and P_2 are the power of a signal at point 1 and 2 respectively.

- 21. Which one of the following media is multidrop?**
- (a) Shielded Twisted pair cable
 (b) Unshielded Twisted pair cable
 (c) Thick Coaxial cable
 (d) Fiber Optic cable

Ans : (c) In a bus network, a single length of wire, cable or optical fiber connects a number of computers. All the computers are connected to a single transmission media normally a thin or thick coaxial cable, in the form of tapings.

- 22. What is the baud rate of the standard 10 Mbps Ethernet?**

- (a) 10 megabaud (b) 20 megabaud
(c) 30 megabaud (d) 40 megabaud

Ans : (b) The Ethernet uses **Manchester encoding**, which means it has two signal periods per bit sent. The data rate of the standard ethernet is 10 Mbps, so the baud rate is twice, or 20 megabaud.

23. At any iteration of simplex method, if $\Delta_j(z_j - C_j)$ corresponding to any non-basic variable x_j is obtained as zero, the solution under the test is

- (a) Degenerate solution
(b) Unbounded solution
(c) Alternative solution
(d) Optimal solution

Ans : (c) Rule: If all $\Delta_j \geq 0$, the solution under the test will be optimal. Alternate optimal solution will exist if any non-basic Δ_j is also zero.

24. A basic feasible solution to a m-origin, n-destination transportation problem is said to be _____ if the number of positive allocations are less than $m+n-1$.

- (a) degenerate (b) non-degenerate
(c) unbounded (d) unbalanced

Ans : (a) Basic feasible solution: Feasible solution to a m-origin and n-destination problem is said to be basic feasible solution if the number of positive allocation are $(m + n - 1)$. If the number of allocations in a basic feasible solution are less than $(m + n - 1)$; it is called **degenerate basic feasible solution** (otherwise non-degenerate).

25. The total transportation cost in an initial basic feasible solution to the following transportation problem using Vogel's Approximation method is

	W 1	W 2	W 3	W 4	W5	Supply
F1	4	2	3	2	6	8
F2	5	4	5	2	1	12
F3	5	5	4	7	3	14
Demand	4	4	6	8	8	

- (a) 76 (b) 80
(c) 90 (d) 96

Ans : (b) Optimality test using stepping stone method. Allocation table is:

	D_1	D_2	D_3	D_4	D_5	Dummy	Supply
S_1	4(4)	2(4)	3	2(_d ₎	6	0	8
S_2	5	4	5(_c ₆)	2(_c ₆)	1	0	12
S_3	6	5	4	7(_c ₂)	3(_c ₈)	0(4)	14
Supply	4	4	6	8	8	4	

The minimum total transportation cost is
 $= 4 \times 4 + 2 \times 4 + 2 \times 8 + 1 \times 4 + 4 \times 6 + 3 \times 4 + 0 \times 4$
 $= 80$

Note: Alternative solution is available with unoccupied cell $S_3D_1=0$, but with the same optimal value.

26. An actor in an animation is a small program invoked _____ per frame to determine the characteristics of some object in the animation.

- (a) once
- (b) twice
- (c) 30 times
- (d) 60 times

Ans : (a) The use of **actors** is a high-level form of procedural control. An actor in an animation is a small program involved once per frame to determine the characteristics of same object in the animation. An actor, in the course of its once-per-frame execution, may send message to other actors to control their behaviors.

27. Bresenham line drawing algorithm is attractive because it used

- (a) Real arithmetic only
- (b) Integer arithmetic only
- (c) Floating point arithmetic
- (d) Real and integer arithmetic

Ans : (b) **Bresenham's algorithm** offers a significant advantage over the other method, as it avoids floating-point arithmetic and rounding. It is a classic example of an incremental algorithm that computes the location of each pixel along the line, based on information about the previous pixel. It uses only integer values and avoids any multiplications.

28. There fresh rate above which a picture stops flickering and fuses into a steady image is called _____.

- (a) Crucial fusion frequency
- (b) Current frequency fusion
- (c) Critical fusion frequency
- (d) Critically diffused frequency

Ans : (c) The **refresh rate of a CRT** is the number of times per second the image is redrawn. It is typically 60 per second for raster displays. As the refresh rate decreases, flicker develops because the eye can no longer integrate the individual light impulses coming from a pixel. The refresh rate above which a pictures stop flickering and fuses into a steady image is called the critical fusion frequency or CFF. The process of fusion is familiar to all of us, it generally occurs whenever we watch television or motion pictures.

29. In homogenous coordinate system (x, y, z) the points with z=0 are called

- (a) Cartesian points
- (b) Parallel points
- (c) Origin point
- (d) Point at infinity

Ans : (d) In homogenous coordinates system (x, y, z) the points with $z = 0$ are called **point at infinity**.

30. If 40 black lines interleaved with 40 white lines can be distinguished across one inch, the resolution is

- (a) 40 line-pairs per inch
- (b) 80 line-pairs per inch
- (c) 1600 lines per inch
- (d) 40 lines per inch

Ans : (a) **Resolution** is defined as the closest spacing at which adjacent black and white lines can be distinguished to observers which implies that horizontal

and vertical resolution may differ. If 40 black lines that are interleaved with 40 white lines can be distinguished across one inch, the resolution is 80 lines per inch, the resolution is 80 lines per inch (also referred to as 40 lines pairs per inch).

- 31. Images tend to be very large collection of data. The size of memory required for a 1024 by 1024 image in which the colour of each pixel is represented by a n -bit number, (in an 8 bit machines) is**
- (a) $n \times 8$ MB (b) $n/8$ MB
(c) $(1024 \times 1024)/8$ MB (d) 1024 MB

Ans : (b)
Size of 1 pixel = n bit = $n/8$ Bytes
Size of 1024×1024 image
= $1024 \times 1024 \times n/8$ Bytes
= $n/8 2^{20}$ Bytes
= $n/8$ MB

- 32. Arrays in C language can have _____ with reference to memory representation.**
- (a) n -subscripts
(b) two-subscripts
(c) only one subscript
(d) three subscripts only

Ans : (c) Every individual element of an array is referenced by appending a subscript in square brackets behind the name.
Array in C can have only one-subscript, but arrays can have arrays as elements, thus supporting multidimensional arrays. This is an example of **orthogonality**.

- 33. Refer the points as listed below:**
- (i) What are the operator precedence rules?
(ii) What are the operator associativity rules?
(iii) What is the order of operand evaluation?
(iv) Are there restrictions on operand evaluation side effects?
- Which of the above must be considered as primary**
- (a) (i), (ii) and (iii)
(b) (i), (iii) and (iv)
(c) (i), (ii) and (iv)
(d) (i), (ii), (iii) and (iv)

Ans : (d) Arithmetic expressions: Unary, binary and ternary operators meaning an operator can have one, two or three operand respectively. Design issues include:

1. What are the operator precedence rules?
2. What are the operator associativity rules?
3. What is the order of operand evaluation?
4. Are there restrictions on operand evaluation side effects?
5. Does the language allow user-defined-operator overloading?
6. What type mixing is allowed in expressions?

- 34. Horn clauses are special kinds of propositions which can be described as**
- (a) Single atomic proposition on left side.
(b) Single or multiple atomic proposition on left side.

- (c) A single atomic proposition on left side and a single atomic proposition on right side.
- (d) A single atomic proposition on left side or an empty left side.

Ans : (d) Horn clauses: These are restricted kind of clauses which must be used to specify-propositions that are used for resolution. These propositions either have a single atomic proposition on the left side or an empty left side.

35. Which of the following is/are the fundamental semantic model (s) of parameter passing?

- (a) in mode
- (b) out mode
- (c) in-out mode
- (d) all of these

Ans : (d) Formal parameters are characterized by one of three distinct semantic models:

1. In mode: They can receive data from corresponding actual parameters.

2. Out mode: They can transmit data to the actual parameter.

3. In-out Mode: They can do the functionally of both IN and OUT mode.

36. The grammar with production rules

$S \rightarrow aSb | SS | \lambda$

generates language L given by:

- (a) $L = \{ w \in \{a,b\}^* \mid n_a(w) = n_b(w) \text{ and } n_a(v) \geq n_b(v) \text{ where } v \text{ is any prefix of } w \}$
- (b) $L = \{ w \in \{a,b\}^* \mid n_a(w) = n_b(w) \text{ and } n_a(v) \leq n_b(v) \text{ where } v \text{ is any prefix of } w \}$
- (c) $L = \{ w \in \{a,b\}^* \mid n_a(w) \neq n_b(w) \text{ and } n_a(v) \geq n_b(v) \text{ where } v \text{ is any prefix of } w \}$
- (d) $L = \{ w \in \{a,b\}^* \mid n_a(w) \neq n_b(w) \text{ and } n_a(v) \leq n_b(v) \text{ where } v \text{ is any prefix of } w \}$

Ans : (a) Given grammar is

$S \rightarrow aSb | SS | \epsilon$

Each S generates equal number of a's and b's.

i.e. $n_a(w) = n_b(w)$.

Since, before each 'b', one 'a', is produced, number of a's will be greater or equal to number of b's in any prefix of string.

Therefore, language is:

$L = \{ w \in \{a,b\}^* \mid n_a(w) = n_b(w) \}$

and $n_a(v) \geq n_b(v)$ where v is any prefix of w }

37. A pushdown automaton $M = (Q, \Sigma, \Gamma, \delta, q_0, Z, F)$ is set to be deterministic subject to which of the following condition (s), for every $q \in Q, a \in \Sigma \cup \{\lambda\}$ and $b \in \Gamma$

S1: $\delta(q, a, b)$ contains at most one element

S2: If $\delta(q, \lambda, b)$ is not empty then $\delta(q, c, b)$ must be empty for every $c \in \Sigma$

- (a) only S1
- (b) only S2
- (c) both S1 and S2
- (d) neither S1 nor S2

Ans:(c) A pushdown automata $M = \{Q, \Sigma, \Gamma, q_0, z, F, \delta\}$ is said to be deterministic if it is an automata as subjected to the restrictions that, for every $q \in Q, a \in \Sigma \cup \{\epsilon\}$ and $b \in \Gamma$,

- (i) $\delta(q, a, b)$ contains at most one element.
- (ii) if $\delta(q, \lambda, b)$ is not empty, then $\delta(q, c, b)$ must be empty for every $c \in \Sigma$.

The first condition states that for any given input symbol and any stack top, at most one move can be made. The second condition states that if a λ -move is possible for some configuration no input consuming alternative should be possible for that configuration.

38. For every context free grammar (G) there exists an algorithm that passes any $w \in L(G)$ in number of steps proportional to

- (a) $\ln |w|$
- (b) $|w|$
- (c) $|w|^2$
- (d) $|w|^3$

Ans : (d) Convert CFG to CNF and then are CYK algorithm which runs in steps proportional to $O(|w|^3)$.

39. Match the following :

List-I

- A. Context sensitive language**
- B. Regular grammar**
- C. Context free grammar**
- D. Unrestricted grammar**

List-II

- 1. Deterministic finite automation**
- 2. Recursive enumerable**
- 3. Recursive language**
- 4. Pushdown automation**

Codes:

	A	B	C	D
(a)	2	1	4	3
(b)	3	4	1	2
(c)	3	1	4	2
(d)	2	4	1	3

Ans : (c) **1.** Regular grammar - Deterministic finite - automation
2. Context - free grammar - pushdown automation
3. Context sensitive language - Recursive language
4. Unrestricted grammar - Recursive enumerable

40. The statements s1 and s2 are given as:

S1: Context sensitive languages are closed under intersection, concatenation, substitution and inverse homomorphism.

S2: Context free languages are closed under complementation, substitution and homomorphism.

Which of the following are correct statements?

- (a) Both S1 and S2 are correct.
- (b) S1 is correct and S2 is not correct
- (c) S1 is not correct and S2 is correct
- (d) Both S1 and S2 are not correct

Ans : (b) **1. Context sensitive languages** are closed under complement, union, intersection, concatenation, substitution, inverse-homomorphism and kleen closure. But are **not closed under** homomorphism property.

2. Context free languages are closed under substitution and homomorphism but are not closed under complement, intersection operation.

41. Which one of the following is not an addressing mode?

- (a) Register indirect
- (b) Auto increment
- (c) Relative indexed
- (d) Immediate operand

Ans : (c) Relative indexed is not an addressing mode.

42. Computers can have instruction formats with

- (a) only two address and three address instructions
- (b) only one address and two address instructions
- (c) only one address, two address and three address instructions
- (d) zero address, one address, two address and three address instructions

Ans : (d) 1. **Stack** operations uses zero-address, that is there is no address field in the computational instructions.

2. One address instructions use an **accumulator** (AC) register for all data manipulation.

3. Two-address instructions are the most common in **commercial computers**.

4. Computers with **three-address instruction** formats can use each address field to specify either a processor register or a memory operand.

43. Which is not a typical program control instruction?

- (a) BR
- (b) JMP
- (c) SHL
- (d) TST

Ans : (c) Typical program control instructions:

Name	Mnemonic
Branch	BR
Jump	JMP
Skip	SKP
Call	CALL
Return	RET
Compare (by subtraction)	CMP
Test (by ANDing)	TST

SHL (shift left) operation is not a typical program control instruction.

44. Interrupt which arises from illegal or erroneous use of an instruction or data is

- (a) Software interrupt
- (b) Internal interrupt
- (c) External interrupt
- (d) All of the above

Ans : (b) 1. **Internal interrupts** arise from illegal or erroneous use of an instruction or data.

Example of conditions that raises internal interrupts are registers overflow, attempt to divide by zero, an invalid operation code, stack overflow and protection violation.

2. External interrupts generally occurs as a result of outside interference. A **software interrupt** often used to implement system call.

45. The simplified function in product of sums of Boolean function $F(W, X, Y, Z) = \Sigma(0, 1, 2, 5, 8, 9, 10)$ is

- (a) $(W'+X')(Y'+Z')(X'+Z)$
- (b) $(W'+X')(Y'+Z')(X'+Z')$

- (c) $(W'+X')(Y'+Z)(X'+Z)$
 (d) $(W'+X')(Y+Z')(X'+Z)$

Ans : (a) Given Boolean function is:

$$f(w, x, y, z) = \sum(0, 1, 2, 5, 8, 9, 10)$$

K-map is:

yz wx				
	00	01	11	10
00	1 0	1 1	0 3	1 2
01	0 4	1 5	0 7	0 6
11	0 12	1 13	0 15	0 14
10	1 8	1 9	0 11	1 10

Product of sum is:

$$f = (\bar{w} + \bar{x}).(\bar{y} + \bar{z}).(\bar{x} + \bar{z})$$

46. Match the following :

List-I

- A. TTL
 B. ECL
 C. MOS
 D. CMOS

List-II

1. High component density
 2. Low power consumption
 3. Evolution of "diode-transistor-logic"
 4. High speed digital circuit

Codes:

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 3 | 2 | 1 | 4 |
| (b) | 1 | 4 | 3 | 2 |
| (c) | 3 | 4 | 1 | 2 |
| (d) | 1 | 2 | 3 | 4 |

Ans : (c) **TTL:** Evolution of diode-transistor-logic

ECL: High speed digital circuits

MOS: High component density

CMOS: Low power consumption

47. Match the following :

List-I

- A. Foreign keys
 B. Private key
 C. Event control action model
 D. Data security

List-II

1. Domain constraint
 2. Referential integrity
 3. Encryption
 4. Trigger

Codes:

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 3 | 2 | 1 | 4 |
| (b) | 2 | 1 | 4 | 3 |
| (c) | 3 | 4 | 1 | 2 |
| (d) | 1 | 2 | 3 | 4 |

Ans : (b) A **foreign key** is an attribute that references to some attribute of another table. The purpose of the foreign key is to ensure referential integrity of the data.

A **trigger** is a special kind of stored procedure that automatically executes when an event occurs in the database server. It's action control model. **Domain constraints** states that the value of every attribute must only belong to the predefined domain.

Encryption is the most effective way to achieve data security.

48. When an array is passed as a parameter to a function which of the following statements is correct?

- (a) The function can change values in the original array.
- (b) The function cannot change values in the original array.
- (c) Results in compilation error
- (d) Results in runtime error.

Ans : (a) Since we are passing array in C, we will pass it by reference. Hence whatever changes we made in called function, that will be reflected in calling function.

49. Suppose you want to delete the name that occurs before "Vivek" in an alphabetical listing. Which of the following data structures shall be most efficient for this operation?

- (a) Circular linked list
- (b) Double linked list
- (c) Linked list
- (d) Dequeue

Ans : (b) We have to search a key and delete the previous node of the node to which key belongs.

In linked list and circular linked list we cannot go to previous node directly.

But in doubly linked list we can access previous node of a given node directly since there is a pointer to previous node in O(1) time.

50. What will be the output of the following segment of the program?

```
main()
{
    char*s="hello world";
    int I=7;
    printf("%s",s);
}
```

- (a) Syntax error
- (b) hello w
- (c) hellow
- (d) o world

Ans : (a) Syntax error.

51. Trace the error:

```
void main ( )
{
    int*b, & a;
    *b=20
    Print("%d, %d", a, *b)
}
```

- (a) No error
- (b) Logical error
- (c) Syntax error
- (d) Semantic error

Ans : (c) There is syntax error 'a' declared as reference but not initialized.

52. Match the following :

List-I

A. calloc

B. free ()

C. malloc ()

D. realloc ()

List-II

1. Frees previously allocated space
2. Modifies previously allocated space
3. Allocates space for array
4. Allocates requested size of space

Codes:

	A	B	C	D
(a)	3	1	4	2
(b)	3	2	1	4
(c)	3	4	1	2
(d)	4	2	3	1

Ans : (a) `calloc ( )`: allocates space for array.
 `free ( )`: frees previously allocated space.
 `malloc ( )`: allocates requested size of space.
 `realloc ( )`: modifies previously allocated space.

53. Binary symmetric channel uses

- (a) Half duplex protocol
- (b) Full duplex protocol
- (c) Bit oriented protocol
- (d) None of the above

Ans : (a) A **binary symmetric** channel is a common communication channel model used in coding theory and information theory. In this model transmitter wishes to send a bit (0 or 1) and receiver receive a bit. At a time only one node can send and other will be in receiving mode. It used half-duplex protocol.

54. Hamming distance between 100101000110 and 110111101101 is

- (a) 3
- (b) 4
- (c) 5
- (d) 6

Ans : (d) The Hamming distance between two strings of equal length is the number of positions at which the corresponding symbols (bits) are different.

Apply bit wise EXOR ($\oplus$) operation:

	1001	0100	0110
$\oplus$	1101	1110	1101
	0100	1010	1011

So, hamming distance is 6 bits.

55. Given code word 1110001010 is to be transmitted with even parity check bit. The encoded word to be transmitted for this code is

- (a) 11100010101
- (b) 11100010100
- (c) 1110001010
- (d) 111000101

Ans : (a) **Even parity** refers to a parity checking mode in asynchronous communication systems in which an extra bit, called a parity bit, is set to one if there is odd number of 1's else 0.

Given code is 1110001010 has odd number of 1's. So, we add 1 to LSB to make even number of 1's in total before send. Therefore, transmitted data would be:

11100010101

56. The number of distinct binary images which can be generated from a given binary image of right $M \times N$ are

- (a) $M+N$
- (b) $M \times N$
- (c) 2^{M+N}
- (d) 2^{MN}

Ans : (d) The number of distinct binary images which can be generate from a given binary image of right $M \times N$ are $2^{M \times N}$.

57. If $f(x, y)$ is a digital image, then x, y and amplitude values of f are

- (a) Finite
- (b) Infinite
- (c) Neither finite nor infinite
- (d) None of the above

Ans : (a) If ' x ', ' y ' and the amplitude values of ' f ' are finite and discrete quantities, then this image is called a digital image. A digital image is composed of a finite number of elements called pixel, each of which has a particular location and value.

58. Consider the following processes with time slice of 4 milliseconds (I/O requests are ignored):

Process	A	B	C	D
Arrival time	0	1	2	3
CPU cycle	8	4	9	5

The average turnaround time of these processes will be

- (a) 19.25 milliseconds
- (b) 18.25 milliseconds
- (c) 19.5 milliseconds
- (d) 18.5 milliseconds

Ans : (b) The system uses Round Robin algorithm with time quantum 4 ms.

Gantt chart is:

A	B	C	D	A	C	D	C
---	---	---	---	---	---	---	---

0 4 8 12 16 20 24 25 26

Average TAT

$$= \frac{(20-0) + (8-1) + (26-2) + (25-3)}{4}$$

$$= \frac{73}{4} = 18.25 \text{ milliseconds}$$

59. A job has four pages A, B, C, D and the main memory has two page frames only. The job needs to process its pages in following order: A B A C A B D B A C D

Assuming that a page interrupt occurs when a new page is brought in the main memory, irrespective of whether the page is swapped out or not. The number of page interrupts in FIFO and LRU page replacement algorithms are

- (a) 9 and 7
- (b) 7 and 6
- (c) 9 and 8
- (d) 8 and 6

Ans : (c)

(i) FIFO algorithm:

A B A C A B D B A C D

A A hit C C B B hit A A D

 B hit B A A D hit D C C

Total number of page faults = 9

(ii) LRU algorithm:

A B A C A B D B A C D

A A hit A hit A D hit A A D

 B hit C hit B B hit B C C

Total number of page faults = 8

60. Suppose S and Q are two semaphores initialized to 1. P1 and P2 are two processes which are sharing resources.

(P1)

wait (S)

wait (Q)

critical section 1;

signal (S);

signal (Q);

Their execution may sometimes lead to an undesirable situation called

(a) Starvation

(b) Race condition

(c) Multithreading

(d) Deadlock

(P2)

wait (Q);

wait (S);

critical section2:

signal (Q);

signal (S);

Ans : (d) If the process P_1 executes wait (S) and preempted then P_2 executes wait (Q) then no further execution of either of the processes is possible. This situation is called **deadlock**.

61. An operating system using banker's algorithm for deadlock avoidance has ten dedicated devices (of same type) and has three processes P1, P2 and P3 with maximum resource requirements of 4, 5 and 8 respectively. There are two states of allocation of devices as follows:

State 1

Processes	P1	P2	P3
Devices allocated	2	3	4

State2

Processes	P1	P2	P3
Devices allocated	0	2	4

Which of the following is correct?

(a) State 1 is unsafe and state 2 is safe.

(b) State 1 is safe and state 2 is unsafe.

(c) Both, state 1 and state 2 are safe.

(d) Both, state 1 and state 2 are unsafe.

Ans : (a) State 1:

Total number of devices allocated
 $= 2 + 4 + 3 = 9$

Number of available
 $= 10 - 9 = 1$

Process	Remaining need
P_1	$4 - 2 = 2$
P_2	$5 - 3 = 2$
P_3	$8 - 4 = 4$

Therefore, no process can be executed.

Hence system is in unsafe state.

State 2:

Total number of device allocated
 $= 0 + 2 + 4 = 6$

Number of available
 $= 10 - 6 = 4$

Process	Remaining need
P_1	$4 - 0 = 4$
P_2	$5 - 2 = 3$
P_3	$8 - 4 = 4$

Therefore, any process can start execution. Hence, system is in safe stable.

62. Let the time taken to switch between user mode and kernel mode of execution be T_1 while time taken to switch between two user processes be T_2 . Which of the following is correct?

- (a) $T_1 < T_2$
- (b) $T_1 > T_2$
- (c) $T_1 = T_2$
- (d) Nothing can be said about the relation between T_1 and T_2 .

Ans : (a) Time taken to switch between two process is large as compared to time taken to switch between Kernel and user mode of execution. When you switch process, you have to do a context switch, save the PCB of previous process, then save registers and then load the PCB of new process and load its registers etc. When you switch between Kernel and user mode of execution, OS has to just change a single bit at hardware level which is a fast operation.

Therefore, $T_1 < T_2$.

63. Working set model is used in memory management to implement the concept of

- (a) Swapping
- (b) Principal of Locality
- (c) Segmentation
- (d) Thrashing

Ans : (b) Working set is a concept which defines the amount of memory that a process requires in a given time interval. The concept is important for effective use of caches, which has principle of locality.

64. A UNIX file system has 1 KB block size and 4-byte disk addresses. What is the maximum file size if the inode contains ten direct block entries, one single indirect block entry, one double indirect block entry and one triple indirect block entry?

- (a) 30 GB
- (b) 64 GB
- (c) 16 GB
- (d) 1 GB

Ans : (c) Maximum file size

$$\begin{aligned}
 &= \left(\frac{\text{Disk block size}}{\text{Direct block address size}} \right)^3 \times \text{Disk block size} \\
 &= \left(\frac{1\text{KB}}{4\text{B}} \right)^3 \times 1\text{K Bytes} \\
 &= (2^8)^3 \times 1\text{K Bytes} \\
 &= 2^{24} \times 2^{10} \text{ bytes} \\
 &= 16 \times 2^{30} \text{ bytes} \\
 &= 16\text{G bytes}
 \end{aligned}$$

65. A thread is usually defined as a light weight process because an Operating System (OS) maintains smaller data structure for a thread than for a process. In relation to this, which of the following statement is correct?

- (a) OS maintains only scheduling and accounting information for each thread.
- (b) OS maintains only CPU registers for each thread.

- (c) OS does not maintain a separate stack for each thread.
- (d) OS does not maintain virtual memory state for each thread.

Ans : (d) OS on per thread basis, maintains only CPU register state and stack. Threads share address space of process. Virtually memory is concerned with processes not with thread.

66. The versions of windows operating system like windows XP and window Vista uses following file system :

- (a) FAT-16
- (b) FAT-32
- (c) NTFS (NT File System)
- (d) All of the above

Ans : (d) Windows XP professional supports the FAT-16, FAT-32 and NTFS file systems. Because NTFS has all the basic capabilities of FAT-16 and FAT-32, with the added advantage of advanced storage feature such as compression, improved security and larger partitions and file sizes. It is a recommended file system for windows vista.

67. Which one of the following is a correct implementation of the meta-predicate "not" in PROLOG (Here G represents a goal)?

- (a) not (G): -!, call (G), fail, not(G)
- (b) not (G): -call (G), !, fail. not(G)
- (c) not (G): -call (G), fail,!, not (G)
- (d) not (G): -call (G), fail, not (G): -!

Ans : (b) By for the most common version of 'not' (or\+) available in PROLOG systems can be implemented as follows:

not (G): - call (G), ! fail
not (G)

Inequality can be implemented using 'not', but is normally implemented as a special case, as follows:

$x \neq x$: - !, fail
 $x \neq y$

If not (G) is called and the call to G fails, then not (G) succeeds. This is correct by the NAF role. If the call to G succeed, cut and fail are called so not (G) fails. This is not part of the NAF role and can cause incorrect behaviour of G contains variables.

68. Which one of the following is not an informed search technique?

- (a) Hill climbing search
- (b) Best first search
- (c) A* search
- (d) Depth first search

Ans : (d) Hill climbing is a mathematical optimization technique which belongs to the family of local search. It is an informed search

A* search is an informed search algorithm used for path-finding and graph traversal.

Informed methods add domain - specific information to select the best path to continue searching along.

Depth first search is not informed search algorithm.

69. If we convert

$\exists u \forall v \forall x \exists y (P(f(u), v, x, y) \rightarrow Q(u, v, y))$ to
 $\forall v \forall x (P(f(a), v, x, g(v, x)) \rightarrow Q(a, v, g(v, x)))$
This process is known as

- (a) Simplification (b) Unification
(c) Skolemization (d) Resolution

Ans : (c) The process of removing all the existential quantifiers from a formula is known as **skolemization**. The resultant formula is skolem normal form that is equivalent to computational complexity of the original.

- 70. Given two jugs of capacities 5 litres and 3 litres with no measuring markers on them. Assume that there is endless supply of water. Then the minimum number of states to measure 4 litres water will be**

- (a) 3 (b) 4 (c) 4 (d) 7

Ans : (d)

Water in	3 Litre Jug	5 Litre Jug
state 1	0	0
state 2	0	5 (fill)
state 3	3	2 (remaining)
state 4	0 (empty)	2
state 5	2	0
state 6	2	5 (fill again)
state 7	3	4 (remaining)

Therefore, minimum number of 7 states are required.

- 71. The map colouring problem can be solved using which of the following technique?**

- (a) Means-end analysis
(b) Constraint satisfaction
(c) AO* search
(d) Breadth first search

Ans : (b) **Constraint satisfaction problems** are mathematical problems defined as a set of objects whose state must satisfy a number of constraints or limitations. Examples of simple problems that can be modeled as a constraint satisfaction - problem include:

1. Eight queens puzzle
2. Map coloring problem
3. Crosswords, Sudoku, other logic puzzles.

- 72. Which of the following is a knowledge representation technique used to represent knowledge about stereotype situation?**

- (a) Semantic network
(b) Frames
(c) Scripts
(d) Conceptual Dependency

Ans : (c) **Knowledge representation** and reasoning (KR) is the field of artificial intelligence (AI) dedicated to representing information about the world in a form that a computer system can utilize to solve complex tasks such as diagnosing a medical condition or having a dialog in a natural language. All of these, in different ways, involve hierarchical representation of data:

1. Lists
2. Trees
3. Semantic networks - nodes and links
4. Schemes - used to represent common sense or stereotyped knowledge.
 - (i) Frames - Describes objects
 - (ii) Scripts - Describes event rather than object. Consist of stereotypically ordered - causal or temporal chain of events.

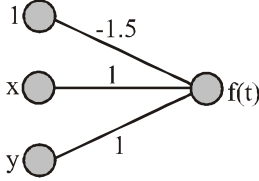
73. A fuzzy set A on R is _____ iff
 $A(\lambda x_1 + (1-\lambda)x_2) \geq \min [A(x_1), A(x_2)]$
 for all $x_1, x_2 \in R$ and all $\lambda \in [0, 1]$, where min denotes the minimum operator.
- (a) Support (b) α -cut
 (c) Convex (d) Concave

Ans : (c) A fuzzy set A defined on R^n is **convex** iff $A(\lambda x_1 + (1-\lambda)x_2) \geq \min (A(x_1), A(x_2))$,
 for all $\lambda \in [0, 1], x_1, x_2 \in R^n$ and all $\alpha \in [0, 1]$, or, equivalently, A is convex if and only if all its α -cuts are convex sets, for any α in the interval $\alpha \in (0, 1]$.

74. If A and B are two fuzzy sets with membership functions
 $\mu_A(x) = \{0.6, 0.5, 0.1, 0.7, 0.8\}$
 $\mu_B(x) = \{0.9, 0.2, 0.6, 0.8, 0.5\}$
 Then the value of $\mu_{A \cup B}(x)$ will be
- (a) $\{0.9, 0.5, 0.6, 0.8, 0.8\}$
 (b) $\{0.6, 0.2, 0.1, 0.7, 0.5\}$
 (c) $\{0.1, 0.5, 0.4, 0.2, 0.2\}$
 (d) $\{0.1, 0.5, 0.4, 0.2, 0.3\}$

Ans : (c) Union operation is fuzzy set takes maximum value between two and its **complement** will be calculated by subtracting each number from '1'.
 Therefore,
 $\mu_{(A \cup B)}(x) = \{0.9, 0.5, 0.6, 0.8, 0.8\}$ and
 $\mu_{(A \cup B)}(x) = \{0.1, 0.5, 0.4, 0.2, 0.2\}$

75. Consider a single perception with weights as given in the following figure :



and $f(t)$ defined as $f(t) = \begin{cases} 1, & t > 0 \\ 0, & t \leq 0 \end{cases}$

The above perception can solve

(a) OR problem (b) And problem
 (c) XOR problem (d) All of these

Ans : (b) Perception after multiplying weights with input will give:
 $\Rightarrow 1(-1.5) + x \cdot 1 + y \cdot 1 = f(t)$
 $\Rightarrow f(t) = x + y - 1.5$
 According to given definition of $f(t)$.
 if $f(t) = 1$, if $t > 0$ else $f(t) = 0$.
 So, truth table is:

x	y	f(t)
0	0	0
0	1	0
1	0	0
1	1	1

e.g. if $x = 0, y = 0$
 $\Rightarrow f(t) = -1.5 < 0$,
 So, $f(t) = 0$
 So, table is truth table of AND.